Reaction Time Combine

Train like a sports scientist and prove which strategy cuts delay.

Win condition: Best median plus most improvement using clean data.

THE SCIENCE YOU ARE USING

From signal to muscle. Reaction time is the delay between seeing a signal and moving. The eye sends a signal to the brain, the brain decides, and nerves fire the muscle. The classic ruler-drop test turns that delay into a distance you can measure.

Median and improvement. One trial is noisy, so you take many and use the median. Then you test whether a strategy, like focusing or warming up, actually improves your number. Clean data and real improvement both score.

HOW TO BUILD AND RUN IT

01 **Run the ruler drop.** A partner drops a meter stick; you catch it. Read the catch distance in cm, then use the cm-to-ms strip to turn it into your reaction time.

02 **Take many trials.** Record at least ten catches. Use the median, not the best single catch.

03 **Pick a strategy.** Choose one strategy to test: focus cue, warm-up, or eyes on the release point.

04 **Retest and compare.** Run another set with the strategy. Compare medians to see if it helped.

05 **Show the data.** Present a clear before-and-after with your median and spread.

Safety: Clear lanes. No collisions. All stations can be seated. Allergy: None.

MATERIALS

Meter sticks	per team
Reaction-time strip (cm to ms)	per team
Pencils	shared
Score sheets	per team
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______

The ONE variable we are testing: ______

Baseline result: ______

After redesign: ______

DEFEND YOUR DESIGN

What evidence made you change your design? ______

If you ran it again, what is the ONE thing you would change? ______

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	**100**